ENTERPRISE SAAS PLATFORM

UNIFIED ARCHITECTURE DOCUMENT

Automotive Material Compliance & IMDS Intelligence Platform

Incorporates: Architecture v1.0 + Gap Analysis v1.1 + Presentation Features + RFI Requirements + iPoint Analysis + Full AI/ML Architecture

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1. Product Research Summary

1.1 Document Analysis Summary

This architecture consolidates findings from all uploaded documents, internet research, and competitive analysis into one definitive reference:

- Nissan RFI v1.9: 15-page requirements specification covering 17 software functions (MDS, SDS, SUI, SCIP, CLP, REACh, RRR) plus 10 system interfaces.
- Marklytics Proposal v4 (5 slides): Competitive analysis of iPoint's 8 core modules, Marklytics base offering (Collect-Analyse-Report-Automate-Integrate), 6 AI innovations, 5 process improvements, competitive benchmarking.
- iPoint Software Analysis: 7 core modules identified — Product Compliance Engine, IMDS Integration Engine, Sustainability/PCF Engine, SCIP Reporting Engine, Supplier Data Management, SAP/ERP Integration, Data Processing & Analytics.
- Gap Analysis v1.1: 14 gaps identified (3 CRITICAL, 5 HIGH, 6 MEDIUM) with full remediation plans.
- IMDS 15.0/15.2 Research: PCF mandatory fields, substance transparency enhancements, chemical classification updates.
- EU Regulatory Landscape 2025–2026: DPP mandates, CBAM financially binding, CSRD reporting, PFAS restriction proposal, new ELV revision.

1.2 What IMDS Platforms Do

The International Material Data System (IMDS) is the automotive industry's global repository for material composition data. Established by BMW, Mercedes-Benz, Ford, GM, Volkswagen, and others, it enables OEMs and suppliers to track every substance in every component of a vehicle. IMDS is operated by DXC Technology and used by over 53 OEMs and 120,000+ supplier companies worldwide.

IMDS platforms like iPoint provide software that sits on top of the raw IMDS system to automate compliance checking, substance analysis, regulatory reporting, and supplier data collection. A car has approximately 10,000 components and each supplier must declare materials through IMDS — processes that are largely manual, complex, and error-prone, creating enormous pressure on the supply chain.

1.3 How Material Compliance Reporting Works

Material compliance in automotive follows a hierarchical data flow model:

- Tier-N suppliers create Material Data Sheets (MDS) documenting the chemical composition of raw materials, semi-components, and components.
- Each MDS is identified by part number and contains substance data with CAS numbers, weight percentages, and regulatory flags (GADSL, SVHC).
- MDS data flows up the supply chain: Tier-2 submits to Tier-1, Tier-1 assembles and submits to OEM.
- OEMs validate MDS against regulation lists (REACH, ELV, GADSL, RoHS) and either accept or reject submissions.
- Accepted data feeds into SCIP notifications (ECHA), REACh registration, CLP labelling, and RRR reporting.

1.4 Nissan-Specific Scale Requirements

Dimension	Scale
Suppliers Managed	4,000+

Material Data Sheets	32,000
SDS PDFs Archived	17,000
BOMs Analysed	500+ (up to 100k parts each)
Languages Supported	26
Global Regions	Europe, Japan, GOM, Americas
SDS Translations per Year	200 mixtures
Aftermarket/Service Parts	Up to 300,000
Supplier Chasing	Automated batch (currently manual email)

1.5 Key Workflows (17 RFI Functions + 6 AI Innovations)

The platform must support all 17 Nissan RFI functions plus Marklytics' 6 proposed AI innovations:

#	Function	Source	Priority
1	MDS Data Collection (direct + indirect parts)	RFI §1	Critical
2	SDS Collection & PDF Archive (17k PDFs, draft/approve workflow)	RFI §2	Critical
3	Safe Use Instructions (SUI) Collection & Validation	RFI §3	High
4	Service Parts MDS Assembly (parent→child propagation)	RFI §4	High
5	Nissan-Built Parts MDS (IMDS trees for site assemblies)	RFI §5	High
6	SVHC Volume Tracking (per legal entity, threshold alerts)	RFI §6	Critical
7	BOM Analysis (100k+ parts, sub-second, historical rules)	RFI §7	Critical
8	Global MDS Data Sharing (EU/JP/Americas, no restrictions)	RFI §8	High
9	SDS Authoring (XML generation, SDS→MDS linkage)	RFI §9	Medium
10	SUI Validation & Publishing (Part + Vehicle level)	RFI §10	Medium
11	Global Substance Analysis (non-EU + EU parts)	RFI §11	High
12	REACh Registration (tonnage tracking, threshold alerts)	RFI §12	High
13	REACh Restriction & Authorisation (Annex XIV/XVII, sunset dates)	RFI §13	High
14	SCIP Identification & Notification (bulk ECHA submission)	RFI §14	Critical
15	CLP Classification & Labelling (26 languages, pictograms)	RFI §15	High
16	Poison Centre Notification (PCN via IUCLID)	RFI §15.8	Medium
17	RRR — Reusability/Recyclability/Recoverability	RFI §16	Medium
AI-1	AI Compliance Prediction (40% fewer rejections)	Presentation	High
AI-2	Smart Material Classification (NLP, 90% less manual entry)	Presentation	High

AI-3	Intelligent Supplier Chasing (risk scoring, 70% faster)	Presentation	High
AI-4	Real-Time Regulatory Alerts (weeks of early warning)	Presentation	High
AI-5	Digital Product Passport (EU Battery/Vehicle regulation-ready)	Presentation	High
AI-6	Predictive Analytics Dashboard (forecasting, resource planning)	Presentation	Medium

2. System Architecture

2.1 Architecture Decision: Modular Monolith + CQRS + Event-Driven

After evaluating domain complexity, team size, and deployment requirements, we recommend a Modular Monolith architecture for the core platform, augmented by event-driven microservices for specific workloads and a CQRS (Command Query Responsibility Segregation) pattern for BOM analysis performance:

- Strong domain boundaries within a single deployable unit, reducing operational complexity.
- Shared database with schema-level isolation per module, ensuring data consistency for compliance-critical operations.
- CQRS pattern: Write Model (PostgreSQL) for MDS CRUD; Read Model (OpenSearch + Redis) for pre-computed compliance projections enabling sub-100ms BOM analysis.
- Event bus (Amazon EventBridge + SQS) for asynchronous workloads that benefit from independent scaling.
- Clear migration path to full microservices in Phase 5+.

2.2 High-Level Architecture Diagram

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┌────────────────────────────────────────────────────────────┐ │                CLIENT TIER
(CDN: CloudFront)                 │ │  Supplier App  │  OEM Portal  │  Admin  │  Mobile
│ └──────────────────────────────┬─────────────────────────────┘
│ ┌──────────────────────────────┴─────────────────────────────┐ │  API GATEWAY (Kong) +
WAF + Zero Trust mTLS                  │
└────────────────────────────┬───────────────────────────────┘
│ ┌──────────────────────────────┴─────────────────────────────┐ │       MODULAR
MONOLITH (NestJS/TypeScript)                │ │  Compliance │ MDS Mgmt │ Supplier │ SDS │
SCIP         │ │  REACH │ CLP │ Sustainability │ Reporting │ RRR          │
└────────────────────────────┬───────────────────────────────┘
│ ┌──────────────────────────────┴─────────────────────────────┐ │           EVENT BUS
(EventBridge + SQS)                    │
└─┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┘   │      │      │      │
│      │      │      │  AI/ML  Rust   Go     Go     Doc   Notif  CQRS  Audit  (Py)
Engine Sync  ERP    Proc   Svc   Proj  Logger        BOM   IMDS   SAP   Λ      SES
Worker DDB
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2.3 Updated Technology Stack (Post Gap Analysis)

Layer	Technology	Rationale
API + Business Logic	NestJS 10 (TypeScript) on ECS Fargate	Excellent for CRUD, auth, DI, module boundaries
Compliance Engine	Rust (WASM or gRPC service)	5–30x performance for 100k+ BOM analysis vs Node.js
IMDS Sync Worker	Go (goroutines) on ECS Task	Superior concurrency for parallel API sync calls
ERP/PLM Connectors	Go (goroutines) on ECS Task	Goroutines excel at parallel SAP RFC/IDoc/OData calls
AI/ML Service	Python FastAPI on ECS	Compliance prediction, NLP extraction, risk scoring
Frontend	Next.js 15+ (Turbopack, App Router, RSC)	SSR/SSG, 26-language i18n, Shadcn/ui + Tailwind
Read Model (CQRS)	OpenSearch + Redis projections	Sub-100ms BOM analysis queries on 100k+ parts

Primary Database	RDS PostgreSQL (Multi-AZ) + RLS	Transactional data, MDS, compliance records
Tenant Model	Hybrid: Shared RLS + Dedicated RDS (Enterprise)	Physical isolation for OEM customers like Nissan
Search Engine	OpenSearch Service	Full-text search across 32k MDS + 17k SDS
Cache	ElastiCache Redis	Session store, CQRS projections, job queues
Object Storage	S3 (Intelligent Tiering)	SDS PDFs, XML files, audit exports with lifecycle policies
Event Bus	EventBridge + SQS	Async event routing between modules
Observability	OpenTelemetry + Grafana (Tempo/Loki/Mimir)	SLO-based alerting, business-level dashboards
Security	Zero Trust + SPIFFE/SPIRE + mTLS	Every call re-validates identity, device posture
CDN	CloudFront	Static assets, global frontend delivery
CI/CD	GitHub Actions + ECR + ECS Blue/Green	Container registry, automated deployment pipeline
IaC	Terraform modules per environment	Reproducible infrastructure, tenant provisioning
Chaos Testing	AWS FIS + Circuit Breakers	Production resilience validation

3. Component Breakdown

3.1 Core Domain Modules (10 Modules)

3.1.1 Compliance Engine (Rust WASM/gRPC)

The central intelligence module that validates material data against regulatory rule sets. Rewritten in Rust for 5–30x throughput on CPU-intensive BOM analysis.

- Rule Evaluation: Validates MDS against current and historical regulation versions. Historical rules preserved for retrospective analysis (RFI §7 requirement).
- SVHC Detection: Flags substances exceeding 0.1% w/w threshold per REACH Article 33.
- PFAS Rule Pack: Explicit PFAS restriction candidate list with per-substance CAS mapping and substitution recommendations (Gap #5).
- BOM Analysis (CQRS): Pre-computed compliance projections in Redis. Write Model (PostgreSQL) for MDS CRUD; Read Model (OpenSearch+Redis) for sub-100ms 100k+ part analysis (Gap #2).
- AI Compliance Prediction: ML models trained on historical rejection patterns predict non-compliance before submission (Presentation AI-1: 40% fewer rejections).
- Nissan Engineering Standard: OEM-specific rules alongside IMDS guidelines, ELV, GADSL (RFI §7).

3.1.2 MDS Management Module

Handles the full lifecycle of Material Data Sheets: creation, import, validation, versioning, and submission.

- IMDS Integration: Bi-directional sync with IMDS (import/export MDS, CAMDS support) via Go workers.
- IMDS 15.0/15.2 PCF Fields: Mandatory pcf_product (kg CO_2e per component/kg) and pcf_transport (kg CO_2e per delivery route) fields. Manufacturing site linkage for PCF provenance (Gap #1).
- Indirect Parts + Materials: Extend MDS collection beyond direct parts to include indirect parts (production/packaging) and raw materials/substances/mixtures (RFI §1.1).
- EEA Import Flag: Track whether product is imported from outside the EEA for REACh analysis (RFI §1.2).
- XML Import from SDS: Import MDS data from SDS XML files generated by SDS authoring tool (RFI §1.2, §2).
- Tree Builder: Visual MDS tree editor for assembling component hierarchies (Vehicle → Component → Sub-component → Material → Substance).
- Service Parts MDS Assembly: Combine existing MDS into service part trees; auto-propagate parent updates to service parts (RFI §4).
- Nissan-Built Parts MDS: Create IMDS trees for assemblies at Nissan sites (aftersales, external contractors, Renault partners) (RFI §5).
- Version Control: Full audit trail with diff comparison between MDS versions.
- Letter of Compliance: Store compliance letters when supplier is unavailable, conforming to Nissan template (RFI §1.3).
- Smart Material Classification (AI): NLP engine auto-extracts and classifies substances from supplier data sheets (Presentation AI-2: 90% less manual entry).

3.1.3 Supplier Management Module

Manages the supplier relationship lifecycle for compliance data collection across 4,000+ suppliers.

- Supplier Portal: Self-service interface for suppliers to submit MDS, SDS, and SUI documents.
- AI-Driven Chasing: Intelligent outreach prioritised by risk score, deadline urgency, and response history. Batch messaging (1 supplier, many parts) (Presentation AI-3: 70% faster response).
- Intelligent Rate Control: Automatic throttling of automated responses to minimise/halt where appropriate (RFI §17.1).
- Historical Communication Archive: Full archive of all supplier communications for reference (RFI §17.1).
- Response Tracking Dashboard: Response rates, overdue submissions, escalation status.
- Management Reports: List suppliers not responded, MDS needing review by expiry date (RFI §1.3).

3.1.4 SDS Management Module

Manages Safety Data Sheets across all regions with multi-language support.

- PDF Archive: Structured storage for 17,000+ SDS PDFs with OCR indexing. 10-year retention per mixture with translations (RFI §2).
- Draft/Approve/Reject Workflow: Supplier uploads → Draft status → Nissan engineer notified → Review → Approve/Reject. Rejected = ‘reviewed pending amendments’ with comments (RFI §2).
- Cross-Region SDS: SDS collected in any region accessible by all regions. Store ‘where mixture is used’ (organisation, department, section), ‘how used’, multiple locations (RFI §2).
- SDS→MDS Linkage: SDS linked to MDS data set. XML file generated per SDS containing substance/mixture contents for MDS import (RFI §2, §9).
- eSDS Support: Store extended Safety Data Sheets with exposure scenarios (REACh >10t/a hazardous chemicals) (RFI §2).
- Workers Instructions: Store workers instructions in various formats, related to each substance/mixture use (RFI §2).
- AI Translation: NLP-assisted translation engine supporting 26 languages (Presentation: 50% cost cut).

3.1.5 SCIP/ECHA Module

Automates SCIP dossier generation and submission to the European Chemicals Agency.

- Dossier Builder: Auto-generates IUCLID i6z format dossiers with substance data, concentration ranges, material identification, TARIC codes.
- TARIC Code Management: Import and maintain customs classification codes per article (RFI §14).
- Bulk Submission: Submit hundreds of SCIP notifications via ECHA’s system-to-system API (RFI §14).
- Simplified SCIP: Perform simplified SCIP notification for different legal entities of registered uses (RFI §14).
- UUID Tracking: Manage ECHA-assigned UUIDs and candidate package list IUCLID references (RFI §14).

3.1.6 REACh Module

Complete REACh registration, restriction, and authorisation support.

- Registration Tracking: List all imported/manufactured substances and mixtures by legal entity with tonnage bands (RFI §12).

- Entity Capture: Track ‘entities using substances’, ‘how product is used’, whether imported or manufactured (RFI §12).
- Volume Monitoring: Alert when substance volumes cross 1t, 10t thresholds. Import tonnage from files (xls, xml) (RFI §12).
- Registration Closure: Enter submission info and close alerts when registration is completed (RFI §12).
- Registered Uses Report: Show registered uses to confirm Nissan use is in line (RFI §12).
- Restriction List: Track Annex XVII restricted substances with restriction dates and restricted uses in articles/mixtures (RFI §13).
- Authorisation Tracking: Track Annex XIV authorised substances with sunset dates, authorised uses (RFI §13).
- SVHC Volume Tracking: Track import/manufacturing volumes per legal entity per SVHC. When total weight exceeds threshold, generate ECHA notification report (RFI §6).
- Direct Material Volumes: Enter/upload direct material volumes used per vehicle for total SVHC content determination (RFI §6).
- Communication Obligation: Flag products subject to REACh communication (SVHC >0.1% w/w) (RFI §7).
- Notification Obligation: Combine SVHC content with import/production volumes to report by legal entity (RFI §7).

3.1.7 CLP Module

Complete Classification, Labelling and Packaging of Substances and Mixtures support.

- Hazard Classification: Identify hazardous substances from SDS Section 2 and non-hazardous mixtures containing hazardous substances from SDS Section 3 (RFI §15.1-2).
- CLP Notification Candidates: Maintain and provide list of CLP notification candidates (RFI §15.4).
- Label Generation: Auto-generate GHS-compliant labels with pictograms, signal words, H/P statements in 26 languages. Support multiple label files per product (standard + small package) (RFI §15.5, 15.7).
- Label Document Upload: Upload additional documentation such as final labels of marketed products (RFI §15.5).
- Search/Display/Retrieve: Full search, display, and download capability for all CLP information (RFI §15.6).
- PCN Support (Gap #12): UFI Generator (16-character unique formula identifiers per mixture per EU country). PCN Dossier Builder (IUCLID i6z with toxicological data). ECHA PCN Portal connector (RFI §15.8).

3.1.8 Sustainability & PCF Module (Gap #1 Remediation)

New module addressing IMDS 15.0/15.2 PCF requirements and upcoming sustainability regulations.

- PCF Data Model: pcf_product (kg CO_2e per component/kg), pcf_transport (kg CO_2e per delivery route), manufacturing_site_id linked to ManufacturingSite entity.
- IMDS 15.x PCF Subscription: Weekly digest of PCF changes in accepted MDS.
- PCF Dashboard: Carbon intensity visualisation per product line, supplier, and region.
- CBAM Module (Gap #4): Track CN/TARIC codes per imported material, embedded emissions (direct + indirect), country of origin, production facility. Calculate CBAM certificates using EU methodology.
- CSRD Data Export: APIs exposing compliance data in ESRS format for corporate sustainability reports. Link SVHC volume tracking to Scope 3 emissions calculations.
- Digital Product Passport (DPP): Generate EU Battery Regulation and Vehicle Regulation-compliant passports from IMDS/compliance data automatically (Presentation AI-5).

- Catena-X EDC Integration (Gap #10): Eclipse Dataspace Connector for sovereign PCF data exchange using Pathfinder framework.

3.1.9 RRR Module (Gap #5 from RFI §16)

Reusability, Recyclability, Recoverability calculation per ELV Directive 2000/53/EC and RRR Directive 2005/64/EC.

- Material Classification Taxonomy: Metallic, polymeric, fluid, electronics, other — with recyclable/recoverable/disposable flags.
- ISO 22628 Calculation Engine: Mass-based recyclability (≥85%) and recoverability (≥95%) rate per vehicle.
- Recycled Content Tracking: % recycled per material (pre/post-consumer, ELV-origin). Aid calculation of 25% recycled plastic target (25% from recycled ELVs).
- RRR Compliance Certificates: Generate per vehicle model for type-approval documentation.
- Critical Raw Materials: Account for recovery of CRMs, plastics, steel, and aluminium.

3.1.10 Reporting & Analytics Module

Executive dashboards and compliance reporting engine.

- Executive Dashboards: Real-time compliance status, supplier response rates, risk heatmaps.
- Predictive Analytics (AI-6): Forecasting deadline clusters, resource needs, and risk trends weeks ahead.
- Standard Reports: Missing MDS, missing SDS, SVHC volume reports, SVHC by legal entity, RRR calculations, tonnage reports (RFI §7).
- Real-Time Regulatory Alerts (AI-4): Continuous monitoring of ECHA, REACH, CLP changes with instant BOM impact analysis.
- BOM List Management: Store and manage BOMs per business function (design-BOM, orderable parts, parts on stock) (RFI §7).
- Export Engine: PDF, Excel, CSV, XML export in compliance-ready formats.

3.2 Async Microservices (Updated per Gap Analysis)

Service	Runtime	Purpose
AI/ML Service	Python FastAPI on ECS	Compliance prediction, NLP substance extraction, supplier risk scoring, material classification
Rust Compliance Engine	Rust gRPC on ECS	BOM analysis, SVHC detection, PFAS scanning, rule evaluation (5–30x faster)
CQRS Projection Worker	Node.js on ECS	Event-driven: MDS change → update OpenSearch+Redis read model
IMDS Sync Worker	Go on ECS Task	Bi-directional IMDS sync with PCF data support (IMDS 15.x)
ERP/PLM Connector	Go on ECS Task	SAP BOM sync (RFC/IDoc/OData), Oracle PLM, volume imports
Document Processor	Lambda (Node.js)	PDF parsing, OCR, SDS extraction, XML generation
Notification Service	Lambda + SES/SNS	Email, in-app, SMS notifications with 26-language templates
Audit Logger	Lambda + DynamoDB Streams	Immutable audit trail, compliance evidence chain

4. Security Architecture (Zero Trust)

The platform implements a Zero Trust security model (Gap #3 remediation), going beyond perimeter security to enforce ‘never trust, always verify’ at every layer:

- Identity: AWS Cognito + SAML 2.0 federation (Azure AD, Okta, ADFS). Short-lived JWTs (15 min access, 7-day refresh) with tenant claims.
- Continuous Authentication: Re-verify session integrity at configurable intervals, not just at login.
- mTLS Everywhere: Mutual TLS for all inter-service communication (not just external APIs).
- SPIFFE/SPIRE: Service identity management for Kubernetes/ECS workload identity.
- Micro-Segmentation: Each module runs in isolated network segments with explicit allow-list policies.
- Device Trust: Certificate-based device binding for OEM portal users.
- RBAC + ABAC: Organisation-scoped permissions (Supplier Admin, Compliance Engineer, OEM Reviewer, Platform Admin) + attribute-based policies for data-level access (region, legal entity, product line).
- MFA: Enforced for all OEM portal users; optional for suppliers.
- Encryption: AES-256 at rest (AWS KMS), TLS 1.3 in transit, VPC private subnets with NAT gateway.
- WAF: AWS WAF with OWASP Top 10 rule set on API Gateway.
- SOC 2 Type II: Architecture designed for compliance from day one.
- GDPR: Data residency controls (EU region); right-to-erasure workflows; DPA management.
- Annual Penetration Testing: Third-party pen tests; automated DAST/SAST in CI pipeline.

5. Data Architecture

5.1 Core Data Schemas (Updated for IMDS 15, PCF, RRR)

5.1.1 Material Data Sheet (MDS)

Field	Type	Description
id	UUID	Primary key
tenant_id	UUID	Organisation scope (RLS enforced)
part_number	VARCHAR(50)	Nissan part number identifier
imds_id	VARCHAR(20)	External IMDS reference ID
supplier_id	UUID	FK to supplier
status	ENUM	DRAFT | SUBMITTED | ACCEPTED | REJECTED
version	INTEGER	Auto-incrementing version number
tree_structure	JSONB	Hierarchical component/material tree
imported_from_eea	BOOLEAN	Whether imported from outside EEA (RFI §1.2)
part_type	ENUM	DIRECT | INDIRECT | MATERIAL | SERVICE_PART | NISSAN_BUILT
parent_mds_id	UUID	FK for service part assembly linkage (RFI §4)
pcf_product	DECIMAL	Product carbon footprint kg CO_2e/kg (IMDS 15)
pcf_transport	DECIMAL	Transport carbon footprint kg CO_2e/route (IMDS 15)
manufacturing_site_id	UUID	FK to ManufacturingSite for PCF provenance
region	ENUM	EU | JP | GOM | AMERICAS
linked_sds_id	UUID	FK to linked SDS record (RFI §2/§9)
letter_of_compliance	TEXT	Stored when supplier unavailable (RFI §1.3)
created_at	TIMESTAMPTZ	Creation timestamp
updated_at	TIMESTAMPTZ	Last modification timestamp

5.1.2 Substance Record (Updated for PFAS)

Field	Type	Description
id	UUID	Primary key
cas_number	VARCHAR(20)	Chemical Abstracts Service identifier
ec_number	VARCHAR(20)	EC/EINECS number
name	VARCHAR(255)	IUPAC or common name
is_svhc	BOOLEAN	On REACH Candidate List
is_pfas	BOOLEAN	Per/polyfluoroalkyl substance flag (Gap #5)
gadsl_status	ENUM	PROHIBITED | DECLARABLE | NONE
reach_annex_xiv	BOOLEAN	Subject to authorisation
reach_annex_xvii	BOOLEAN	Subject to restriction
clp_classification	JSONB	GHS hazard classes and categories
sunset_date	DATE	REACh authorisation sunset date
pfas_substitution	TEXT	Recommended substitute (AI-powered)

rrr_classification	ENUM	METALLIC | POLYMERIC | FLUID | ELECTRONICS | OTHER
recyclable	BOOLEAN	Recyclable flag for RRR calculation
recycled_content_pct	DECIMAL	% recycled content (pre/post-consumer)

5.2 Multi-Tenant Architecture (Hybrid Model — Gap #8)

- Starter/Professional tiers: Shared DB + RLS (Row-Level Security) on PostgreSQL.
- Enterprise tier (Nissan): Dedicated RDS instance per tenant for physical data isolation.
- TenantRouter Middleware: Resolves database connection string per request based on tenant configuration.
- Tenant Hierarchy: Platform → Organisation (OEM/Supplier) → Business Unit → User, supporting Nissan's 4-region structure.

5.3 Document Storage Architecture

Document Type	Storage	Indexing	Retention
SDS PDFs	S3 Intelligent Tiering	OpenSearch (OCR text)	10 years (RFI §2)
MDS XML Exports	S3 Standard-IA	PostgreSQL metadata	Indefinite
SUI Documents	S3 Standard	OpenSearch (text)	Vehicle lifecycle
CLP Labels	S3 Standard	PostgreSQL metadata	Product lifecycle
Workers Instructions	S3 Standard	OpenSearch (text)	Per substance use
SCIP Dossiers (i6z)	S3 Standard	PostgreSQL metadata	Indefinite
PCN Dossiers (i6z)	S3 Standard	PostgreSQL metadata	Indefinite
Letters of Compliance	S3 Standard	PostgreSQL metadata	Nissan template
Audit Logs	DynamoDB + S3 Glacier	DynamoDB partition key	7 years minimum
DPP Documents	S3 Standard	PostgreSQL + OpenSearch	Product lifecycle
BOM Imports (CSV/XML)	S3 Standard-IA	PostgreSQL metadata	5 years

6. Integration Architecture (RFI §6 Complete)

The platform integrates with all 10 system interfaces specified in the Nissan RFI:

#	System	Protocol	Direction	Data Exchanged
1	Nissan Global BOM	SFTP + REST	Bi-directional	Production BOM, orderable parts, parts on stock, design BOM (RFI §6.1)
2	IMDS (DXC Technology)	SOAP/REST API	Bi-directional	MDS import/export, PCF data, CAMDS, validation (Go worker)
3	SAP ERP	RFC/IDoc/ OData	Inbound	Production BOMs, volumes, stock lists (Go worker)
4	Oracle PLM/PDM	REST API	Inbound	Design BOMs, part metadata, engineering changes
5	ECHA SCIP	IUCLID i6z / REST	Outbound	SCIP dossiers, UUID management, bulk notifications
6	ECHA PCN Portal	IUCLID i6z / REST	Outbound	PCN dossiers with UFI, toxicological data
7	InfoDyne / SDScom	XML / SFTP	Inbound	SDS XML files, substance data (RFI §6.3–4)
8	QuickFDS (P2 Website)	REST API	Outbound	Published SDS and SUI documents (RFI §6.10)
9	Catena-X / DPP	REST (EDC)	Outbound	PCF exchange, Digital Product Passports
10	File Transfer (SFTP)	SFTP / S3 Upload	Inbound	BOM files, volume data, tonnage updates (RFI §6.9)

Additional integration requirements from RFI:

- SUI Upload Interface: Upload list of SVHCs with respective Safe Use Instructions (RFI §6.2).
- InfoDyne Interface 2: Upload files generated by InfoDyne software (labels, PDF files, SDS summaries) linked to Nissan part numbers (RFI §6.5).
- Volume Retrieval: Retrieve volume information per legal entity for parts and substances (RFI §6.6).
- SI Units: Use defined S.I. units throughout, harmonised to avoid unnecessary conversions (RFI §6.7).
- Nissan Data Feed: Accept number of parts per legal entity + import volumes per legal entity for substances and preparations (RFI §6.8).
- Data Export APIs: Allow extracting data for use in other Nissan systems via webservices (RFI §17.2).
- Multi-Lingual Storage: English UI; content stored and translated in 26 languages per Annex 1 (RFI §17.3).

6.1 Webhook & Event Architecture

The platform exposes configurable webhooks for real-time integration:

- mds.submitted / mds.accepted / mds.rejected — MDS lifecycle events
- scip.submitted / scip.confirmed — ECHA submission status
- regulation.updated — SVHC additions, GADSL changes, CLP updates, PFAS changes
- supplier.response_received / supplier.overdue — Supplier chasing events
- bom.analysis_complete — Large BOM analysis completion
- pcf.updated — PCF data changes from IMDS 15.x sync

Webhook payloads signed with HMAC-SHA256. Retry policy: exponential backoff, 3 attempts, dead-letter queue.

6.2 REST API Design

- Base URL: https://api.marklytics.io/v1
- Authentication: Bearer JWT or HMAC-signed API key
- Rate Limits: Starter 100/min, Professional 1000/min, Enterprise 10,000/min
- Pagination: Cursor-based for large datasets
- Batch Operations: POST /v1/mds/batch (up to 500 per request)
- Async Jobs: Long-running operations return job_id with polling endpoint

7. UI Architecture

7.1 Portal Structure

Portal	Users	Key Functions
Supplier Portal	4,000+ supplier companies	Submit MDS/SDS/SUI, view requests, track status, upload documents, respond to chasing
OEM Portal	Nissan compliance engineers	Review submissions, BOM analysis, regulations, SCIP/REACH/CLP, SVHC volumes, RRR
Admin Portal	Platform administrators	Tenant management, user provisioning, system config, billing
Compliance Officer Portal	Senior compliance staff	Executive dashboards, audit reports, regulatory forecasting, DPP management, PCF analytics

7.2 Key Page Designs

OEM Dashboard

- KPI cards: Total MDS coverage %, overdue submissions, SVHC alerts, SCIP pending, PCF completeness.
- Compliance heatmap: Visual status across product lines with PFAS highlighting.
- Deadline timeline: Upcoming regulatory deadlines and submission due dates.
- Supplier response funnel: Requested → In Progress → Submitted → Accepted.
- AI Predictive Compliance: Risk trajectory forecasting over next 12 months (Presentation AI-6).

MDS Submission Workflow (Supplier Portal)

- Step 1: Select part number or scan BOM upload.
- Step 2: Build material tree (drag-and-drop hierarchy builder).
- Step 3: Enter substance data (CAS auto-lookup, SVHC/PFAS flagging, AI classification).
- Step 4: Attach supporting documents (SDS PDF, SUI, certificates, labels).
- Step 5: Validate against compliance rules (real-time feedback with AI prediction).
- Step 6: Submit for OEM review.

BOM Analysis (OEM Portal)

- Upload BOM (CSV/XML/SAP sync) or select from stored BOMs (design-BOM, orderable, stock) (RFI §7).
- Real-time analysis via CQRS read model: sub-100ms for 100k+ parts.
- Results: filterable table with RAG status per part, PFAS flags, PCF data.
- Drill-down to substance level with regulation-specific flags.
- Historical analysis: re-analyse against rules in force at any point in time (RFI §7).

8. Observability & Operations

8.1 Observability Stack (Gap #8 Remediation)

- Instrumentation: OpenTelemetry (OTel) across all services (traces, metrics, logs).
- Backend: Grafana Cloud or self-hosted Grafana + Tempo (traces) + Loki (logs) + Mimir (metrics).
- SLO-Based Alerting: Error budgets per service (e.g., 99.9% MDS submission success rate). Alert on budget burn rate, not thresholds.
- Business Dashboards: MDS submission latency, compliance check throughput, SCIP success rate, supplier response times.
- Correlation IDs: Flow across entire event chain — API request → event bus → worker → external API.
- SLOs: 99.9% uptime; <200ms p95 API response; <100ms BOM analysis (CQRS); <5s full re-analysis.

8.2 FinOps Strategy (Gap #9)

- Reserved Instances: RDS PostgreSQL + ElastiCache Redis (1-year reserved, ~40% saving).
- Spot Instances: BOM analysis workers, document processing Lambdas, AI/ML batch inference.
- S3 Intelligent Tiering: Automatic lifecycle for 17k+ SDS PDFs (hot → warm → cold → Glacier).
- Right-Sizing: Monthly review via AWS Compute Optimizer.
- Cost Allocation Tags: Tag all resources by tenant, module, environment.

8.3 Chaos Engineering & Resilience (Gap #14)

- AWS Fault Injection Simulator (FIS) for controlled chaos in staging.
- Test: RDS failover during BOM analysis, IMDS API timeout, S3 unavailability, ECHA submission failure.
- Circuit Breakers on all external API calls (IMDS, ECHA, SAP).
- RTO: <15min. RPO: <1min for primary database.

8.4 AI/ML Governance (Gap #13)

- EU AI Act Classification: Compliance prediction = likely high-risk; NLP extraction = limited risk.
- Model Cards: Document training data, performance metrics, bias assessments per model.
- Explainability: Predictions show contributing factors (substances, regulations, confidence score).
- Human-in-the-Loop: AI predictions advisory only; human makes final accept/reject decisions.
- Model Monitoring: Detect accuracy drift, trigger retraining alerts.

9. AI/ML Architecture & Intelligence Platform

Marklytics differentiates from legacy tools like iPoint through a purpose-built AI/ML layer that transforms material compliance from reactive data entry into predictive, automated intelligence. This section details the full AI architecture powering the 6 AI innovations referenced throughout this document.

9.1 AI Architecture Overview

The AI subsystem runs as an independent Python FastAPI service (services/ai-ml/) communicating with the NestJS monolith via gRPC and async event queues. This decoupled design allows independent scaling of GPU/compute-intensive workloads and independent model deployment cycles.

```
AI/ML Architecture — High-Level Data Flow
┌──────────────────────────────────────────────────────────────────────┐  │  DATA SOURCES
 │ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐  │ │MDS Submit│
│SDS PDFs  │ │Supplier  │ │Regulatory│ │Historical│  │ │ │Events    │ │& XMLs    │
│Responses │ │Feeds     │ │Outcomes  │ │ │ └────────┘ └──────────┘ └──────────┘
└──────────┘ └──────────┘ │ │
└──────────────────────────────────────────────────────┘ │ │       ▼
│ │ ┌──────────────────────────────────────────────────────────┐ │ │ │
FEATURE STORE  (Redis + PostgreSQL)        │        │ │ │  Material embeddings │
Supplier risk scores │ BOM graphs │        │ │
└──────────────────────────────────────────────────────────┘         │ │
▼                                          │ │
┌──────────────────────────────────────────────────────────┐         │ │ │
AI/ML SERVICE  (Python FastAPI)            │        │ │ │ ┌──────────┐ ┌──────────┐
┌──────────┐ ┌──────────┐ │ │ │ │ │Compliance│ │Material  │ │Intelligent│
│Regulatory│ │ │ │ │ │Predictor │ │Classifier│ │Chaser    │ │Monitor   │ │
│ │ │ │(AI-1) │ │(AI-2) │ │(AI-3) │ │(AI-4) │ │ │ │ │
└──────────┘ └──────────┘ └──────────┘ └──────────┘ │ │ │
┌──────────┐ ┌──────────┐ ┌──────────┐ │ │ │ │ │DPP │ ┌──────────┐
│Translation│ │Anomaly │ │ │ │ │ │Generator │ │Analytics │ │Predictive│
│Detector │ │ │ │ │(AI-5) │ │(AI-6) │ │(NLP) │ │Engine │
│ │ │ └──────────┘ └──────────┘ └──────────┘ │ │(Quality) │ │
└──────────────────────────────────────────────────────────┘ │ │
▼                                          │ │
┌──────────────────────────────────────────────────────────┐         │ │ │
MODEL REGISTRY  (MLflow + S3)              │        │ │ │  Model versions │ A/B
experiments │ Model cards │ Metrics │        │ │
└──────────────────────────────────────────────────────────┘         │
└──────────────────────────────────────────────────────────────────────┘
```

9.2 AI Innovation Models — Detailed Specifications

Each of the 6 Marklytics AI innovations is backed by specific ML models, training pipelines, and measurable KPIs:

AI Innovation	Model Architecture	Training Data	Key Metrics	Deployment
AI-1: Compliance Prediction	XGBoost ensemble + LSTM temporal model	Historical MDS submissions, rejection reasons, regulatory updates (100k+ records)	Precision: 92%, Recall: 88%, F1: 0.90 target (40% fewer rejections)	Real-time inference via gRPC, <200ms latency
AI-2: Material Classification	Fine-tuned multilingual BERT + NER	SDS PDFs, CAS database, SVHC/GADSL	Classification accuracy: 95%+, NER F1: 0.93	Batch + on-demand, GPU-accelerated for

	pipeline	lists, supplier data sheets (50k+ documents)	(90% less manual entry)	PDF processing
AI-3: Intelligent Chasing	Multi-armed bandit + reinforcement learning	Supplier response history, email open rates, deadline patterns (2+ years of interactions)	Response rate: +35%, mean response time: -70% (70% faster responses)	Scheduled batch scoring + event-triggered re-ranking
AI-4: Regulatory Alerts	Transformer-based change detection + knowledge graph	ECHA, REACh, CLP, GADSL regulatory feeds, legal gazette RSS, EU Official Journal	Alert precision: 90%+, BOM impact latency: <30min (weeks of early warning)	Continuous streaming via Kafka consumers
AI-5: DPP Generator	Template engine + LLM summarisation (GPT-4/Claude API)	IMDS data, PCF calculations, supply chain mapping, EU Battery Regulation schema	Schema compliance: 100%, data completeness: 98%+ per DPP	On-demand generation, API-callable
AI-6: Predictive Analytics	Prophet time series + Random Forest resource model	Submission volumes, deadline clusters, seasonal patterns, team capacity data	Forecast MAPE: <15%, resource prediction accuracy: 85%+	Nightly batch retrain, dashboard refresh every 4h

9.3 ML Pipeline & MLOps Infrastructure

The platform implements a full MLOps lifecycle ensuring reproducible training, safe deployment, and continuous monitoring:

Training Pipeline

- Data Ingestion: EventBridge triggers capture MDS submission outcomes, SDS extraction results, and supplier interaction logs into S3 data lake (Parquet format).
- Feature Engineering: Dedicated feature store (Redis for online features, PostgreSQL for offline features) computes 200+ features across material composition, supplier reliability, regulatory proximity, and temporal patterns.
- Training Orchestration: AWS SageMaker Pipelines run scheduled retraining (nightly for AI-1/AI-3, weekly for AI-2/AI-6, on-trigger for AI-4). Each run produces versioned model artifacts stored in S3.
- Experiment Tracking: MLflow tracks all experiments with hyperparameter configs, training metrics, dataset versions, and model lineage. Every model is tagged with a reproducibility hash.
- Validation Gates: Automated model validation compares new model performance against production baseline. Models must exceed minimum thresholds (F1 > 0.85, latency < 500ms) before promotion.

Model Registry & Versioning

- MLflow Model Registry: Central repository for all model versions with stage transitions (Staging → Canary → Production → Archived).
- Model Cards: Every model version includes a standardised card documenting: training data summary, performance metrics, bias assessment, known limitations, EU AI Act risk classification, and intended use scope.
- A/B Testing: Shadow deployment infrastructure allows new models to run in parallel with production, comparing predictions without affecting users. Automatic promotion after 7-day validation window.
- Rollback: One-click rollback to any previous model version with automatic feature store compatibility verification.

Model Serving & Inference

- Online Inference: FastAPI service with gRPC endpoints for low-latency predictions (<200ms p99). Auto-scaling via ECS Fargate with GPU support for NLP models.
- Batch Inference: SageMaker Batch Transform for bulk processing (nightly compliance scoring of entire BOM trees, weekly supplier risk re-ranking).
- Edge Caching: Redis caches frequent prediction results (same material + regulation combination) with 1-hour TTL, reducing inference load by ~60%.
- Model Warm-Up: Pre-loaded models on service start prevent cold-start latency spikes. Health checks verify model availability before accepting traffic.

9.4 AI Data Pipeline Architecture

Data flows from operational systems through transformation layers into the AI subsystem:

Pipeline Stage	Technology	Description	SLA
Ingestion	EventBridge + SQS	Real-time event capture from MDS/SDS/SUI submissions, supplier responses, regulatory feed updates	<1s event delivery
Raw Storage	S3 (Parquet/JSON)	Immutable raw data lake with Hive-compatible partitioning by date/tenant/type	99.999% durability
Transformation	AWS Glue + dbt	Data cleaning, normalisation, deduplication; substance name resolution; CAS number validation	Nightly T+4h completion
Feature Store	Redis (online) + PostgreSQL (offline)	Pre-computed features: material embeddings, supplier scores, regulatory proximity vectors, BOM graph metrics	<10ms online, <1s offline
Training Data	SageMaker Feature Store	Point-in-time correct training datasets preventing data leakage; versioned with DVC	Reproducible to commit hash
Inference Input	gRPC protobuf	Strongly typed inference requests with feature vector validation	<50ms feature assembly
Feedback Loop	EventBridge + SQS	Human decisions (accept/reject/modify) fed back as labelled data for retraining	Continuous capture

9.5 NLP & Document Intelligence

The NLP subsystem powers multiple AI innovations and is the backbone of automated data extraction:

SDS/PDF Extraction Pipeline (AI-2 Core)

- OCR Layer: AWS Textract for scanned PDFs with confidence scoring. Fallback to Tesseract OCR for edge cases.
- Layout Detection: Custom LayoutLM model trained on 5,000+ SDS documents to identify sections (composition tables, hazard statements, regulatory references).
- Entity Extraction: Fine-tuned SpaCy NER model extracts CAS numbers, substance names, concentration ranges, hazard codes (H/P statements), regulatory references.
- Table Extraction: Specialised table detection model for SDS Section 3 (composition) tables. Handles merged cells, multi-page tables, and inconsistent formatting.
- Validation: Extracted data cross-referenced against CAS registry, SVHC candidate list, GADSL, and IMDS basic substance database. Confidence scores per field (>0.95 auto-accepted, 0.80–0.95 flagged for review, <0.80 manual entry required).

Translation Engine

- 26-Language Support: Neural machine translation (NMT) for SDS content using domain-adapted MarianMT models.
- Chemistry-Aware Translation: Custom terminology dictionaries for chemical names, regulatory terms, and IMDS-specific vocabulary to prevent mistranslation.
- Translation Memory: Fuzzy matching against previously translated segments (85%+ match threshold) for consistency and cost reduction (50% translation cost reduction — Presentation metric).

LLM Integration (GenAI Layer)

- DPP Narrative Generation (AI-5): Anthropic Claude/OpenAI GPT-4 API generates human-readable sustainability narratives for Digital Product Passports from structured compliance data.
- Regulatory Summarisation (AI-4): LLM-powered summarisation of regulatory changes from ECHA Official Journal, distilling multi-page legal text into actionable impact assessments.
- Intelligent Query Assistant: Natural language interface for compliance queries (e.g., 'Which suppliers have SVHC substances above 0.1% in components for Qashqai?'). Translates to structured database queries via text-to-SQL.
- Guardrails: All LLM outputs pass through a validation layer checking factual accuracy against the compliance database. Hallucination detection via cross-reference scoring. No LLM output is auto-committed — always human-reviewed.

9.6 AI/ML Technology Stack

Layer	Technology	Purpose
ML Framework	scikit-learn, XGBoost, PyTorch	Classical ML models (AI-1, AI-3, AI-6) + deep learning (AI-2, AI-4)
NLP	SpaCy, Hugging Face Transformers, LayoutLM	NER, document classification, SDS layout parsing

LLM Provider	Anthropic Claude API / OpenAI GPT-4 API	DPP narrative generation, regulatory summarisation, query assistant
Translation	MarianMT (Hugging Face), custom fine-tuned	26-language SDS translation with chemistry domain adaptation
OCR	AWS Textract + Tesseract (fallback)	Scanned SDS/PDF digitisation with confidence scoring
Feature Store	Redis (online) + PostgreSQL (offline)	Real-time feature serving (<10ms) + historical feature storage
Experiment Tracking	MLflow	Model versioning, A/B experiments, model registry, model cards
Training Pipeline	AWS SageMaker Pipelines	Automated retraining, hyperparameter tuning, distributed training
Serving	FastAPI + gRPC + ECS Fargate (GPU)	Low-latency inference (<200ms), auto-scaling, health monitoring
Data Pipeline	EventBridge + SQS + AWS Glue + dbt	Real-time ingestion, batch transformation, feature engineering
Monitoring	Evidently AI + Grafana	Model drift detection, data quality monitoring, performance dashboards
Knowledge Graph	Neo4j	Regulatory relationship mapping, substance-regulation linkage, BOM traversal
Vector Store	OpenSearch (k-NN plugin)	Material similarity search, SDS duplicate detection, regulatory semantic search
Time Series	Prophet + Grafana	Demand forecasting, deadline prediction, seasonal pattern analysis (AI-6)

9.7 AI Governance & EU AI Act Compliance (Gap #13)

All AI capabilities are governed under a formal AI Governance Framework aligned with the EU AI Act (effective August 2025):

AI Capability	EU AI Act Risk Level	Governance Controls
AI-1: Compliance Prediction	High Risk (health/safety decision support)	Human-in-the-loop mandatory; predictions advisory only; full explainability via SHAP values; quarterly bias audits; model cards required
AI-2: Material Classification	Limited Risk	Transparency obligations; confidence thresholds with manual fallback; training data provenance documentation
AI-3: Intelligent Chasing	Minimal Risk	Rate limiting controls; supplier opt-out mechanism; communication audit trail
AI-4: Regulatory Alerts	Limited Risk	Source attribution mandatory; false positive monitoring; human editorial review for high-impact alerts
AI-5: DPP Generator	High Risk (regulatory document generation)	LLM output validation against compliance database; human approval before issuance; full audit trail; no auto-commit
AI-6: Predictive Analytics	Minimal Risk	Forecast confidence intervals displayed; model performance dashboards; anomaly flagging
NLP Translation	Limited Risk	Quality scoring per translation; chemistry term dictionary validation; human review for safety-critical content
Query Assistant	Limited Risk	Text-to-SQL validation; no direct database mutations;

		response grounding in source data

Explainability (XAI) Framework

- SHAP Values: Every compliance prediction (AI-1) includes feature importance breakdown showing which substances, concentrations, regulations, and historical patterns contributed to the risk score.
- LIME: Local interpretable explanations for material classification (AI-2) showing which SDS text segments drove the classification decision.
- Attention Visualisation: For transformer-based models (AI-4), attention heat maps show which parts of regulatory text triggered alerts.
- Confidence Calibration: All models output calibrated probability scores (Platt scaling) so 90% confidence truly means 90% accuracy.

Continuous Monitoring & Drift Detection

- Data Drift: Evidently AI monitors input feature distributions. Alert if KL-divergence exceeds threshold (indicates training data no longer representative).
- Concept Drift: Track prediction-vs-outcome accuracy over rolling 30-day windows. Trigger automatic retraining if F1 drops below 0.85.
- Fairness Metrics: Monitor prediction accuracy across supplier tiers, regions, and company sizes to detect bias. Disparate impact ratio must stay above 0.8.
- Performance SLAs: Real-time dashboards in Grafana track inference latency (p50, p95, p99), throughput, error rates, and model version health.

10. Repository Structure

Turborepo monorepo with updated service structure reflecting Gap Analysis recommendations:

```
saas-imds-platform/ ├── apps/ │   ├── web/                     # Next.js 15+ frontend │
├── api/                     # NestJS modular monolith │   └── mobile/
# React Native ├── packages/ │   ├── shared-types/            # TypeScript shared types
│   ├── ui/                      # Shared React component library │   ├── compliance-
rules/         # Regulation rule packs (REACH, GADSL, ELV, PFAS) │   └── api-client/
# Auto-generated typed API client ├── services/ │   ├── ai-ml/                  #
Python FastAPI ML service │   │   ├── models/              # ML model definitions (AI-1
through AI-6) │   │   ├── pipelines/           # SageMaker training pipelines │   │
├── features/            # Feature store definitions │   │   ├── nlp/                #
NLP/NER/translation engines │   │   ├── inference/           # gRPC serving endpoints │
│   └── monitoring/          # Drift detection, Evidently configs │   ├── rust-
compliance/          # Rust BOM engine (WASM/gRPC) │   ├── cqrs-projector/          #
CQRS read model projection worker │   ├── doc-processor/          # Lambda: PDF/OCR/XML
processing │   ├── imds-sync/              # Go: IMDS bi-directional sync + PCF │   ├──
notification/           # Lambda: Email/SMS/In-app │   └── audit-logger/          #
Lambda: Immutable audit trail ├── connectors/ │   ├── sap-erp/                # Go: SAP
BOM sync connector │   ├── oracle-plm/              # Go: Oracle PLM integration │   ├──
echa-scip/              # ECHA SCIP + PCN submission │   ├── imds-api/                #
DXC IMDS API client library │   ├── infodyne/               # InfoDyne/SDScom/QuickFDS
│   ├── catena-x-edc/           # Eclipse Dataspace Connector │   ├── nissan-global-
bom/       # Nissan Global BOM interface │   └── cbam-csrd/               # CBAM
certificate + CSRD export ├── infra/ │   ├── terraform/                # IaC modules +
environments │   ├── kubernetes/              # Kustomize manifests │   └── docker/
# Dockerfiles per service ├── tests/ │   ├── e2e/                      # Playwright E2E
tests │   ├── integration/             # API integration tests │   ├── load/
# K6 load tests (100k BOM) │   ├── compliance/              # Regulation rule validation
│   └── chaos/                   # AWS FIS chaos experiments └── docs/
# ADRs, API specs, runbooks
```

11. SaaS Business Model

11.1 Subscription Tiers

Dimension	Starter	Professional	Enterprise
Target Customer	Tier-2/3 suppliers	Tier-1 suppliers	OEMs & large Tier-1s
Users	Up to 10	Up to 50	Unlimited
Suppliers Managed	Up to 100	Up to 1,000	Unlimited
MDS Records	Up to 1,000	Up to 10,000	Unlimited
SDS PDF Storage	5 GB	50 GB	Unlimited
BOM Analysis	Up to 1,000 parts	Up to 50,000 parts	100,000+ parts (CQRS)
Compliance Rules	GADSL + ELV	Full (REACH, SCIP, CLP, PFAS)	Full + Custom + OEM rules
AI Features	Not included	Predictive compliance	Full AI suite (all 6 innovations)
IMDS Integration	Manual import/export	Automated sync	Real-time bi-directional + PCF
SCIP Submission	Manual	Semi-automated	Fully automated bulk
ERP Integration	CSV import only	SAP connector	Full ERP/PLM suite
Languages	English only	6 EU languages	26 languages
DPP / Catena-X	Not included	Basic DPP	Full Catena-X + EDC
Sustainability	Not included	PCF dashboard	Full PCF + CBAM + CSRD
Database	Shared (RLS)	Shared (RLS)	Dedicated RDS instance
Support	Email (48hr)	Priority (8hr)	Dedicated CSM (4hr)
SSO	Not included	SAML 2.0	SAML + SCIM provisioning
Indicative Price	£500/mo	£3,000/mo	£15,000+/mo

12. Implementation Roadmap (Revised)

Updated 24-month roadmap incorporating all Gap Analysis re-phasing recommendations:

Phase 1 — Core Platform (Months 1–4)

Foundation, security, PCF data model, and compliance engine core.

- AWS infrastructure (Terraform IaC): VPC, RDS Multi-AZ, ECS, S3, CloudFront.
- NestJS modular monolith scaffold with Zero Trust security from day one (Gap #3).
- Authentication: Cognito + SAML SSO + RBAC/ABAC + mTLS + SPIFFE/SPIRE.
- CQRS pattern for BOM analysis: Write (PostgreSQL) + Read (OpenSearch + Redis) (Gap #2).
- IMDS 15.x PCF data model: pcf_product, pcf_transport, manufacturing_site fields (Gap #1).
- Compliance Engine v1 (Rust): GADSL + ELV + PFAS rule packs (Gap #5).
- Core data models: Organisation, Supplier, MDS (with indirect parts, EEA flag, PCF), Substance (with PFAS), BOM.
- Next.js 15+ frontend shell with sidebar navigation and role-based routing.
- FinOps strategy embedded from start: Reserved Instances, cost allocation tags (Gap #9).
- CI/CD pipeline: GitHub Actions → ECR → ECS Blue/Green deployment.

Milestone: Internal alpha with MDS entry, PCF support, PFAS scanning, CQRS BOM analysis.

Phase 2 — Supplier Portal (Months 5–8)

Supplier-facing workflows, Go workers, and observability.

- Supplier self-registration and onboarding flow.
- MDS submission wizard: tree builder, substance entry, AI material classification (AI-2).
- Indirect parts and materials MDS collection (RFI §1.1).
- SDS collection: PDF upload, draft/approve/reject workflow, cross-region access, eSDS (RFI §2).
- SUI collection and validation workflow with MDS linkage (RFI §3).
- Go workers for IMDS sync (bi-directional + PCF) and ERP connectors (Gap #6).
- AI supplier chasing engine: risk scoring, batch messaging, rate control (AI-3, RFI §17.1).
- Letter of compliance storage when supplier unavailable (RFI §1.3).
- OpenTelemetry + Grafana observability stack (Gap #8).
- Notification service: 26-language email templates, in-app alerts, digest summaries.

Milestone: Live supplier portal handling MDS/SDS/SUI for pilot group with AI chasing.

Phase 3 — OEM Portal (Months 9–12)

Full OEM compliance, PCN, RRR, chaos testing, enterprise DB.

- OEM dashboard: KPIs, compliance heatmap, deadline timeline, supplier funnel, PCF analytics.
- Rust compliance engine: full rule packs (REACH, SCIP, CLP, RoHS, TSCA, PFAS) (Gap #6).
- 100k+ BOM analysis via CQRS read model: sub-100ms response.
- Service Parts MDS assembly with parent→child propagation (RFI §4).
- Nissan-Built Parts MDS creation (RFI §5).

- SVHC volume tracking per legal entity with threshold alerts (RFI §6).
- REACh module: registration, restriction (Annex XVII), authorisation (Annex XIV) (RFI §12–13).
- CLP module with PCN/IUCLID support: UFI generator, PCN dossier builder (Gap #12, RFI §15).
- RRR calculation engine: ISO 22628, material classification, recycled content tracking (RFI §16).
- Database-per-tenant option for Enterprise tier (Gap #8).
- Chaos engineering: AWS FIS, circuit breakers, RTO/RPO testing (Gap #14).
- BOM list management per business function (design-BOM, orderable, stock) (RFI §7).
- Historical rules retention + retrospective analysis (RFI §7).
- Global MDS data sharing across EU/JP/Americas/GOM with no restrictions (RFI §8).

Milestone: Full OEM portal covering all 17 Nissan RFI requirements.

Phase 4 — Integrations & AI (Months 13–18)

Enterprise connectors, full AI suite, sustainability modules.

- SAP ERP connector (Go): real-time BOM sync via RFC/IDoc/OData (RFI §6.1).
- ECHA SCIP connector: automated dossier generation and bulk submission (RFI §14).
- IMDS bi-directional sync: real-time automated MDS exchange with PCF data.
- InfoDyne/SDScom/QuickFDS integration (RFI §6.3–5, 6.10).
- Full AI/ML service: compliance prediction (AI-1), NLP extraction (AI-2), regulatory alerts (AI-4).
- SDS translation engine: AI-assisted 26-language translation (RFI §2, Presentation).
- SDS authoring: XML generation, SDS→MDS linkage (RFI §9).
- SUI validation and publishing at Part + Vehicle level (RFI §10).
- CBAM module: embedded emissions tracking, CBAM certificates (Gap #4).
- CSRD data export in ESRS format (Gap #4).
- Catena-X EDC connector + DPP generator (Gap #10, Presentation AI-5).
- AI governance framework: EU AI Act compliance, model cards, explainability (Gap #13).
- Public REST API + webhook system for customer integrations.

Milestone: Enterprise-ready with full integration suite, AI capabilities, and sustainability modules.

Phase 5 — Enterprise Scaling (Months 19–24)

Global scaling, advanced analytics, and market readiness.

- Multi-region deployment: Read replicas in ap-northeast-1 (Japan) and us-east-1 (Americas) (Gap #11).
- Active-active with CRDTs for MDS data, GDPR data residency enforcement.
- Advanced predictive analytics: regulatory impact simulation, resource planning (AI-6).
- Global substance analysis for non-EU + EU parts (RFI §11).
- Multi-tenant marketplace: third-party regulation rule pack publishing.
- SOC 2 Type II certification and annual penetration testing.
- Performance targets: <200ms p95 API, <100ms BOM (CQRS), 99.9% uptime SLA.

Milestone: Production-grade enterprise SaaS deployable across global automotive supply chains.

13. Presentation Feature Cross-Reference

Verification that every feature from the Marklytics Nissan presentations is reflected in this architecture:

13.1 Presentation Slide Features → Architecture Mapping

Presentation Feature	Architecture Section	Status
iPoint 8 Modules Analysis	§1.1 Document Analysis	INCLUDED
Marklytics Base: Collect-Analyse-Report-Automate-Integrate	§3 Component Breakdown (all 10 modules)	INCLUDED
AI-1: AI Compliance Prediction (40% fewer rejections)	§3.1.1 Compliance Engine + §3.2 AI/ML Service	INCLUDED
AI-2: Smart Material Classification (90% less manual entry)	§3.1.2 MDS Module + §3.2 AI/ML Service	INCLUDED
AI-3: Intelligent Supplier Chasing (70% faster)	§3.1.3 Supplier Module + RFI §17.1 rate control	INCLUDED
AI-4: Real-Time Regulatory Alerts	§3.1.10 Reporting Module + §8.1 Observability	INCLUDED
AI-5: Digital Product Passport	§3.1.8 Sustainability Module + §6 Catena-X EDC	INCLUDED
AI-6: Predictive Analytics Dashboard	§3.1.10 Reporting Module + §7 UI Architecture	INCLUDED
Process: 70% time saved supplier collection	§3.1.3 AI-Driven Chasing + batch messaging	INCLUDED
Process: 60% faster compliance analysis	§3.1.1 Rust Engine + CQRS + AI prediction	INCLUDED
Process: 80% less effort SCIP reporting	§3.1.5 SCIP Module + automated dossier generation	INCLUDED
Process: 50% cost cut SDS translation	§3.1.4 SDS Module + AI translation (26 languages)	INCLUDED
Process: Weeks ahead regulatory monitoring	§3.1.10 Real-Time Regulatory Alerts (AI-4)	INCLUDED
Competitive: AI-Native (vs iPoint Limited)	§2.3 Full AI/ML stack + §8.4 AI Governance	INCLUDED
Competitive: Built-In DPP (vs iPoint Limited)	§3.1.8 Sustainability Module + Catena-X EDC	INCLUDED
Competitive: Core Predictive Forecasting	§3.1.10 Predictive Analytics + AI-6	INCLUDED
Competitive: Modern UX (vs iPoint Basic)	§7 UI Architecture + Next.js 15 + Shadcn/ui	INCLUDED
Scale: 4,000 suppliers, 32k MDS, 26 languages	§1.4 Scale Requirements + all modules	INCLUDED

13.2 iPoint Software Analysis → Architecture Mapping

iPoint Module (from Analysis Doc)	Marklytics Architecture Section	Status
Module 1: Product Compliance Engine	§3.1.1 Compliance Engine (Rust, superior to iPoint)	EXCEEDS
Module 2: IMDS Integration Engine	§3.1.2 MDS Module + Go IMDS Sync	EXCEEDS

	Worker	
Module 3: Sustainability / PCF Engine	§3.1.8 Sustainability Module (PCF + CBAM + CSRD + DPP)	EXCEEDS
Module 4: SCIP Reporting Engine	§3.1.5 SCIP/ECHA Module	MATCHES
Module 5: Supplier Data Management	§3.1.3 Supplier Module (AI-driven, superior)	EXCEEDS
Module 6: SAP / ERP Integration	§6 Integration Architecture (Go connectors)	MATCHES
Module 7: Data Processing & Analytics	§3.1.10 Reporting + §3.2 AI/ML Service	EXCEEDS
iPoint SRD: FR-1 to FR-8	All 10 core modules + async services	FULLY COVERED
iPoint Future Features to Beat	AI prediction, auto-classification, DPP, risk scoring, regulatory updates	ALL INCLUDED

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